USC Dornsife
Dana and David Dornsife
College of Letters, Arts and Sciences
Spatial Sciences Institute

SSCI 220L, Spatial Data Collection Using Drones

Syllabus

Units: 4

Term: Fall 2025
Lecture, Day — Time: Tue/Thu, 9:00-10:20am
Lab, Day — Time: Mon, 8:00-9:50am

Location:
Lecture: SOS B43
Lab: SOS B43

Instructor: Yi Qi, Ph.D
Office: AHF B55J
Office Hours: TBD
Contact Info: yi.qi@usc.edu, 213-821-1589

Library Help: Andy Rutkowski
Office: LIPA B40-A
Office Hours: By Appointment
Contact Info: arutkows@usc.edu

IT Help: Spatial Support
Contact Info: spatial_support@usc.edu

Course Scope and Purpose

Geographic Information Systems (GIS) and imagery collected from unoccupied aerial systems (UASs) or drones present researchers an unparalleled opportunity to study the earth's built and natural systems and the human impact on these systems in very high detail. The recent advent of stable and maneuverable unoccupied aerial systems (e.g., quadcopters and octocopters vs fixed-wing systems), formal regulations by the FAA (in the United States) and the dramatic reduction in production costs, has enabled the rapid adoption over the past five years of UAS by the consumer market. Under the supervision of faculty who are licensed FAA remote system pilots, students will develop the requisite knowledge and practical skills to conduct remote sensing operations via sourcing, analyzing, and producing GIS and simulation-based projects with UAS-derived data. As a recurring theme, the course will use examples from the domain of security and safety of human populations.

This course is a required course for the B.S. in Human Security and Geospatial Intelligence. It serves as an elective course for the B.S. in GeoDesign, the B.S. in Global GeoDesign, and the B.S. in Environmental Studies. Additionally, it serves as an elective course for minors in Human Security and Geospatial Intelligence or Spatial Studies.

All course materials will be organized through Brightspace. The main theoretical concepts will be provided through course notes and assigned readings. Hands-on practical labs will use various software products accessible over the Internet or the SSI Server. Assignments will give students an opportunity to internalize and apply the concepts and theory learned from readings. Some assignments require student interaction, which will benefit all students.

Learning Objectives

On completion of this course, students should be able to:

- Specify fitness-for-purpose criteria (i.e., use cases) and apply them to the evaluation of geospatial data for specific applications.
- Demonstrate an understanding of conceptual foundations of remote sensing, focused primarily on imagery data, with an additional focus on UAS-derived products.
- Describe the methods to collect and process UAS-derived imagery.
- Use GIS and modeling to process, exhibit, and analyze imagery datasets.

Students may vary in their competency levels on these abilities. You can expect to acquire these abilities only if you honor all course policies, attend classes regularly, complete all assigned work in good faith and on time, and meet all other course expectations of you as a student.

Prerequisite(s): None
Co-requisite(s): None
Recommended Preparation: None

Class Conduct

This content is protected and may not be shared, uploaded, or distributed.

Harassment, sexual misconduct, interpersonal violence, and stalking are not tolerated by the university. All faculty and most staff are considered Responsible Employees by the university and must forward all information they receive about these types of situations to the Title IX Coordinator. The Title IX Coordinator is responsible for assisting students with supportive accommodations, including academic accommodations, as well as investigating these incidents if the reporting student wants an investigation. The Title IX office is also responsible for coordinating supportive measures for transgender and non-binary students such as faculty notifications, and more. If you need supportive accommodations you may contact the Title IX Coordinator directly (titleix@usc.edu or 213-821-8298) without sharing any personal information with me. If you would like to speak with a confidential counselor, Relationship and Sexual Violence Prevention Services (RSVP) provides 24/7 confidential support for students (213-740-9355 (WELL); press 0 after hours).

Diversity and Inclusion – It is my intent that students from all diverse backgrounds and perspectives be well served by this course, that students' learning needs be addressed both in and out of class, and that the diversity that students bring to this class be viewed as a resource, strength, and benefit. It is my intent to present materials and activities that are respectful to everyone, and you are also expected to respect of others regardless of their race, ethnicity, gender identity and expressions, cultural beliefs, religion, sexual orientation, national origin, age, abilities, ideas and perspectives, or socioeconomic status. Your suggestions are encouraged and appreciated. Feel free to let me know ways to improve the effectiveness of the course for you personally or for other students.

Course Structure

This is a four-credit course comprised of lectures (two per week) and lab session (one per week). The lecture/learning sessions will discuss various aspects of spatial data collection approaches, remote sensing concepts, UAS platforms, UAS-based sensors, UAS data collection, and the hardware and software systems used to investigate these processes. The weekly lab meetings or UAS data collections are designed to introduce the student to the tools of remote sensing inquiry and to provide practical experience in implementing these tools to explore various problems within the framework of the scientific method. The lecture and lab sessions are designed to complement each other to provide the student with sound theoretical reasoning and the technical skills to investigate various physical attributes. The lab assignments will be graded and returned. It is required that you register for both the lecture and one laboratory session for this course.

Please note that all course materials and correspondence will be posted on the course Brightspace website. As a registered student you will find this course available for you to access on the first day of classes.

Technological and Communication Requirements

ArcGIS and drone imagery processing software are provided online via the SSI Server; hence, you do not need to install it on your own computer. Instead, every student must have the

This content is protected and may not be shared, uploaded, or distributed.

following technology requirements:

- A computer with a fast Internet connection.
- An up-to-date web browser to access the Server

A limited number of computers with all the necessary software is available in the SSI Suite (AHF B55) during regular business hours, Monday through Friday 9 am to 5 pm. To reserve a computer, please use this link: https://calendly.com/usc-ssi/the-ssi-suite-ahf-b55-student-computers-1?month=2024-09. These computers are available to any student in an SSCI or GSEC course and can be used as a resource if you experience difficulties in accessing the SSI server or using the GIS software on your personal computer.

If a student does not have access to any of these, please speak with the instructor at the start of the semester. Also, see the USC ITS Student Toolkit here: https://keepteaching.usc.edu/students/student-toolkit/

Brightspace – This course will utilize the Brightspace learning management system which allows students to access course content, upload assignments, participate in discussion forms, among other learning experiences. The Brightspace platform provides flexibility in the learning experience where students can participate in the course residentially or remotely, synchronously (meeting together at the same time) or asynchronously (accessing videos and course content outside of class).

SSI Server and Tech Support – This course utilizes the SSI Server which is a virtual desktop giving access to many different professional software. If you are unable to connect to the server or experience any type of technical issues, send an email using your USC account to SSI Tech Support at spatial_support@usc.edu, making sure to copy (cc) me on the email.

Communications – All materials to be handed in will be submitted via Brightspace. It is each student's responsibility to stay informed about what is going on in our course. In addition to email about time-sensitive topics, any important announcements will be posted on the Announcement page in Brightspace. Be sure to check these each time you log onto Brightspace. I will send via email through Brightspace any notices that are time sensitive. Please be sure that you read as soon as possible all email sent from Brightspace or from me. Do not ignore course email until the day before assignments are due. Also double check to be sure that email sent from the USC Brightspace account does not go into your junk mail! The instructor will endeavor to respond to all email within 24 hours of receipt, aiming for no more than 72 hours delay. In the rare case that an instructor is off-line for an extended period of time, an announcement will be posted to the class Brightspace site. Due to the synchronous and asynchronous nature of this course, it is each student's responsibility to stay informed and connected with others in our course. In addition to email, students are expected to login to Brightspace regularly to check for announcements.

This content is protected and may not be shared, uploaded, or distributed.

Discussion forums – On the Brightspace site, I will post a series of discussion threads relevant to various sections of the course. Discussions provide a key means for student-to-student discussion and collaboration that can replicate the face-to-face contact you may have experienced in traditional classrooms. Here students can provide support to each other while working on your assignments, sharing hints and helpful tips, as you would in a classroom laboratory. Please post your questions about assignments there, as you would ask them publicly in the classroom. I monitor the discussion threads and offer comments when necessary, but more importantly, consider the discussion board a key way to connect with your classmates and share your discoveries.

Required Readings and Supplementary Materials

The required textbooks for this course are:

- Brown, C. and C. Harder. 2017. *The ArcGIS Book – 2nd Edition*. Washington, DC: New America. Esri Press, Redlands, CA. ISBN: 9781589484870 (no cost; provided in Brightspace).
- Brown, C. and C. Harder. 2016. *The ArcGIS Imagery Book*. Washington, DC: New America. Esri Press, Redlands, CA. ISBN: 9781589484627 (no cost; provided in Brightspace).
- Canada Centre for Remote Sensing (CCRS). 2000. *Fundamentals of Remote Sensing* (no cost; provided in Brightspace).
- Jensen, John R, 2017. *Drone Aerial Photography and Videography: Data Collection and Image Interpretation*. Amazon Kindle store or Apple iBooks store. NOTE: The Amazon Kindle store version is recommended as it can be viewed on either the Microsoft Windows or the Apple operating systems.

Supplemental Readings – Additional readings will be used in course as directed or posted to Brightspace.

Description and Assessment of Assignments

There are several different kinds of assignment with at least one due weekly. These are described in the Weekly Folders on Brightspace. Careful planning and a serious, consistent commitment will be required for a student to successfully navigate the various deliverables in this course throughout the semester. Due dates are shown in the course schedule below.

Lab Assignments – 5 worth a total of points. As the lab portion of the class, students will be doing lab work based on GNSS data, UAS imagery data, or data collected from another source (such as satellite imagery). Students may talk with each other in class to complete these exercises & tutorials but will need to do the labs on their own and submit independent written work.

Midterm – 1 worth 15 points. There will be one midterm administered in class which will cover information presented this far as well as assigned readings.

This content is protected and may not be shared, uploaded, or distributed.

Reading Responses – 5 worth a total of 20 points. During most weeks, each student will complete assigned readings and submit a report on Brightspace. The report includes what the student agrees/disagrees with, what critiques the student may have, or links to other relevant materials (websites, videos, etc.).

Class Interaction – Various worth a total of 5 points. Student participation and involvement is critical for the success of this course. Each student must actively participate in class discussions, labs, and field-based work. Your participation should:

- Demonstrate that you have read and understood the assigned texts and articles
- Help move the class conversation forward by offering your unique/relevant perspective
- Take an active role in field-based work.

Final Exam – 1 worth 20 points. There will be one final exam administered in class which will cover information presented throughout the course. The exam will be administered during exam week.

Grading Breakdown

Careful planning and a serious, consistent commitment will be required for you to successfully navigate the various deliverables in this and other SSI courses. The table below summarizes the SSCI 220L course assignments and their point distribution.

Assignments	Number	Points Each	Total Points
Lab Assignments	5	8	40
Midterm	1	15	15
Reading Responses	5	4	20
Class Interaction (field work)	1	5	5
Final Exam	1	20	20
Totals	**13**		**100**

Grading Scale

Assignments in this and other SSCI courses, are graded on the letter grade scale where A is exemplary, B is very good, C is satisfactory, D is unsatisfactory, and F needs improvement. Final grades use the same letter grade scale with C being the minimum passing grade for credit at the graduate level. The grading scale follows:

A	> 93 points		B-	80-82 points		D+	67-69 points
A-	90-92 points		C+	77-79 points		D	63-66 points
B+	87-89 points		C	73-76 points		D-	60-62 points
B	83-86 points		C-	70-72 points		F	<60 points

This content is protected and may not be shared, uploaded, or distributed.

Assignment Submission Policy

Unless otherwise noted, assignments must be submitted via Brightspace by the due dates specified in the Course Schedule below and on the assignment instructions.
Unless otherwise noted, all Reading Assignments and Tutorials are *due by 11:59 pm Pacific Time (PT) on the deadline*. Lab components have different due dates as indicated on the Course Schedule below. Your attention to on-time assignment submission is essential if I am to meet my goal to return comments on your submitted assignments before the next one is due. Sometimes this is impossible, so I will post a notice on anticipated delays if needed.

Strict penalties apply for late assignments as follows:

- All assignments will be penalized 25% points/day up to FOUR days late. No points will be given for submissions more than FOUR days late.
- Additionally, no written work will be accepted for grading after 5 pm PT on the last day of classes.

Any exceptions to these turn-in assignments are only made by me in coordination with individual students. An example of an exception would be a student's illness or injury that reasonably prohibits course involvement/participation.

SSI Policy on the Creation of Original Work and Use of Generative AI

All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. Students may not have another person or entity complete any substantive portion of an assignment or reuse work prepare for courses without obtaining written permission from the instructor(s). Developing strong competencies in research, writing, and the technical execution of geospatial technologies are foundational to SSI academic programs that are designed to prepare you for success in the workplace. Therefore, using generative AI tools – unless explicitly specified otherwise – is strictly prohibited in this course, will be identified as plagiarism, and will be reported to the Office of Academic Integrity.

Grading Timeline

My goal is to provide grading and feedback on each course assignment no later than two weeks after the assignment was submitted.

Learning Experience Evaluations

Please note Learning Experience Evaluations for the course take place at the end of the semester and are facilitated by the University. These evaluations provide an important review of student experiences in the course.

Course Content Distribution and Synchronous Session Recordings Policies

USC has policies that prohibit recording and distribution of any synchronous and asynchronous course content outside of the learning environment.

Recording a university class without the express permission of the instructor and announcement to the class, or unless conducted pursuant to an Office of Accessibility Services (OSAS) accommodation. Recording can inhibit free discussion in the future, and thus infringe on

This content is protected and may not be shared, uploaded, or distributed.

the academic freedom of other students as well as the instructor. (Living our Unifying Values: The USC Student Handbook, page 13).

Distribution or use of notes, recordings, exams, or other intellectual property, based on university classes or lectures without the express permission of the instructor for purposed other than individual or group study is prohibited. This includes but is not limited to providing materials for distribution by services publishing course materials. This restriction on unauthorized use also applies to all information, which has been distributed to students or in any way has been displayed for use in relationship to the class, whether obtained in class, via email, on the internet, or via any other media. (Living our Unifying Values: The USC Student Handbook, page 13).

Course Schedule: A Weekly Breakdown

	Topic	Reading and Assignments	Deliverables
Module 1: UAS Basics			
Week 1 8/25	Course Overview Intro to GIS and Spatial Data	Review Course Syllabus Brown (2016) Chap. 1 Brown (2017) Chap. 1 Jensen (2017) Chap. 1 CCRS (2000) ToC	
Week 2 9/1 *Monday, 9/1 is university holiday **(Labor Day)**	UAS Anatomy	Lecture Slides Reading Response 1	
Week 3 9/8	UAS Flight Basics (Part I)	Jensen (2017) Chap. 7 Jansen (2017) Chap. 4, Sections 7-8 Lab 1 (9/8)	Reading Response 1 (9/12)
Week 4 9/15	UAS Flight Basics (Part II)	Reading Response 2	Lab 1 (9/19)
Module 2: Foundation in Spatial Data			

This content is protected and may not be shared, uploaded, or distributed.

Week 5 9/22	Intro to Spatial Data Collection and Remote Sensing	Brown (2016) Chap. 2 Brown (2017) Chap. 2 CCRS (2000) Chap. 1 Lab 2 (9/22)	Reading Response 2 (9/26)
Week 6 9/29	Defining Imagery and Intro to sUAS	Brown (2016) Chap. 3 Jensen (2017) Chap. 2 Reading Response 3	Lab 2 (10/3)
Week 7 10/6 *10/9-10/10 is university holiday (Fall Recess)	GNSS Coordinates and Coordinate-Based Referencing	Clark (2020) Chaps. 4 and 7 Supplemental Reading	Reading Response 3 (10/10)
Module 3: Perceiving Imagery			
Week 8 10/13	Special Plan Self-study	Jensen (2017) Chap. 3 Supplemental Reading on Photogrammetry	Midterm (10/23)
Week 9 **10/20**	Perceiving the Imperceptible (Sensors - Part I)	Brown (2016) Chap. 4 Jensen (2017) Chap. 4, Sections 1-6 and Section 11 Lab 3 (10/20)	
Week 10 10/27	Perceiving the Imperceptible (Sensors - Part II)	CCRS (2000) Chap 2 CCRS (2000) Chap 3 Reading Response 4	Lab 3 (10/31)
Week 11 11/3	Turning Imagery into Information (Part I)	Brown (2016) Chap. 5 Jensen (2017) Chap. 5, Sections 1- 5 Lab 4 (11/3)	Reading Response 4 (11/7)
Week 12 11/10 *Tuesday, 11/11 is university holiday (Veterans Day)	Turning Imagery into Information (Part II)	Jenson (2017) Chap. 5, Sections 6-13 Reading Response 5	Lab 4 (11/14)

This content is protected and may not be shared, uploaded, or distributed.

Module 4: Future in Imagery World			
Week 13 11/17	3D Imagery and Temporal Change	Brown (2016) Chap. 6 Jensen (2017) Chap. 6, Sections 1-6 CCRS (2000) Chap 4, Sections 4.1-4.5 Brown (2016) Chap. 7 Jensen (2017) Chap. 6, Sections 7-15 CCRS (2000) Chap 4, Sections 4.6-4.9 Lab 5 (11/17)	Reading Response 5 (11/21)
Week 14 11/24 *11/26-11/28 is a university holiday (Thanksgiving)	The Big Data and Converging Technologies	Brown (2016) Chap. 8 Jensen (2017) Chap 4, Sections 9-10 Jensen (2017) Chap. 8 and 9 Supplemental Reading	
Week 15 12/1	Summary of Imagery Applications and Products	Read Brown (2016) Chap. 9 Course summary and final exam study guide	Lab 5 (12/3)
Final Exams 12/10-12/17	Final Exam Date Thursday, December 11, 11:00 am- 1:00 pm PST		

Statement on Academic Conduct and Support Systems

Academic Integrity:

The University of Southern California is a learning community committed to developing successful scholars and researchers dedicated to the pursuit of knowledge and the dissemination of ideas. Academic misconduct, which includes any act of dishonesty in the production or submission of academic work, comprises the integrity of the person who commits the act and can impugn the perceived integrity of the entire university community. It stands in opposition to the university's mission to research, educate, and contribute productively to our community and the world.

All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. You may not submit work written by others or "recycle" work prepared for other courses without obtaining written permission from the instructor(s).

Other violations of academic integrity include, but are not limited to, cheating, plagiarism, fabrication (e.g., falsifying data), collusion, knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage.

The impact of academic dishonesty is far-reaching and is considered a serious offense against the university. All incidences of academic misconduct will be reported to the Office of Academic

This content is protected and may not be shared, uploaded, or distributed.

Integrity and could result in outcomes such as failure on the assignment, failure in the course, suspension, or even expulsion from the university.

For more information about academic integrity see the student handbook or the Office of Academic Integrity's website, and university policies on Research and Scholarship Misconduct.

Please ask your instructor if you are unsure what constitutes unauthorized assistance on an exam or assignment, or what information requires citation and/or attribution.

Students and Disability Accommodations:

USC welcomes students with disabilities into all of the University's educational programs. The Office of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

Support Systems:

Counseling and Mental Health - (213) 740-9355 – 24/7 on call
Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

988 Suicide and Crisis Lifeline - 988 for both calls and text messages – 24/7 on call
The 988 Suicide and Crisis Lifeline (formerly known as the National Suicide Prevention Lifeline) provides free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week, across the United States. The Lifeline is comprised of a national network of over 200 local crisis centers, combining custom local care and resources with national standards and best practices. The new, shorter phone number makes it easier for people to remember and access mental health crisis services (though the previous 1 (800) 273-8255 number will continue to function indefinitely) and represents a continued commitment to those in crisis.

Relationship and Sexual Violence Prevention Services (RSVP) - (213) 740-9355(WELL) – 24/7 on call
Free and confidential therapy services, workshops, and training for situations related to gender- and power-based harm (including sexual assault, intimate partner violence, and stalking).

Office for Equity, Equal Opportunity, and Title IX (EEO-TIX) - (213) 740-5086

This content is protected and may not be shared, uploaded, or distributed.

Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

Reporting Incidents of Bias or Harassment - (213) 740-5086 or (213) 821-8298
Avenue to report incidents of bias, hate crimes, and microaggressions to the Office for Equity, Equal Opportunity, and Title for appropriate investigation, supportive measures, and response.

The Office of Student Accessibility Services (OSAS) - (213) 740-0776
OSAS ensures equal access for students with disabilities through providing academic accommodations and auxiliary aids in accordance with federal laws and university policy.

USC Campus Support and Intervention - (213) 740-0411
Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

Culture Journey - (213) 740-2101
Information on events, programs and training, the Provost's Diversity and Inclusion Council, Diversity Liaisons for each academic school, chronology, participation, and various resources for students.

USC Emergency - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call
Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety - UPC: (213) 740-6000, HSC: (323) 442-1200 – 24/7 on call
Non-emergency assistance or information.

Office of the Ombuds - (213) 821-9556 (UPC) / (323-442-0382 (HSC)
A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

Occupational Therapy Faculty Practice - (323) 442-2850 or otfp@med.usc.edu
Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.

This content is protected and may not be shared, uploaded, or distributed.